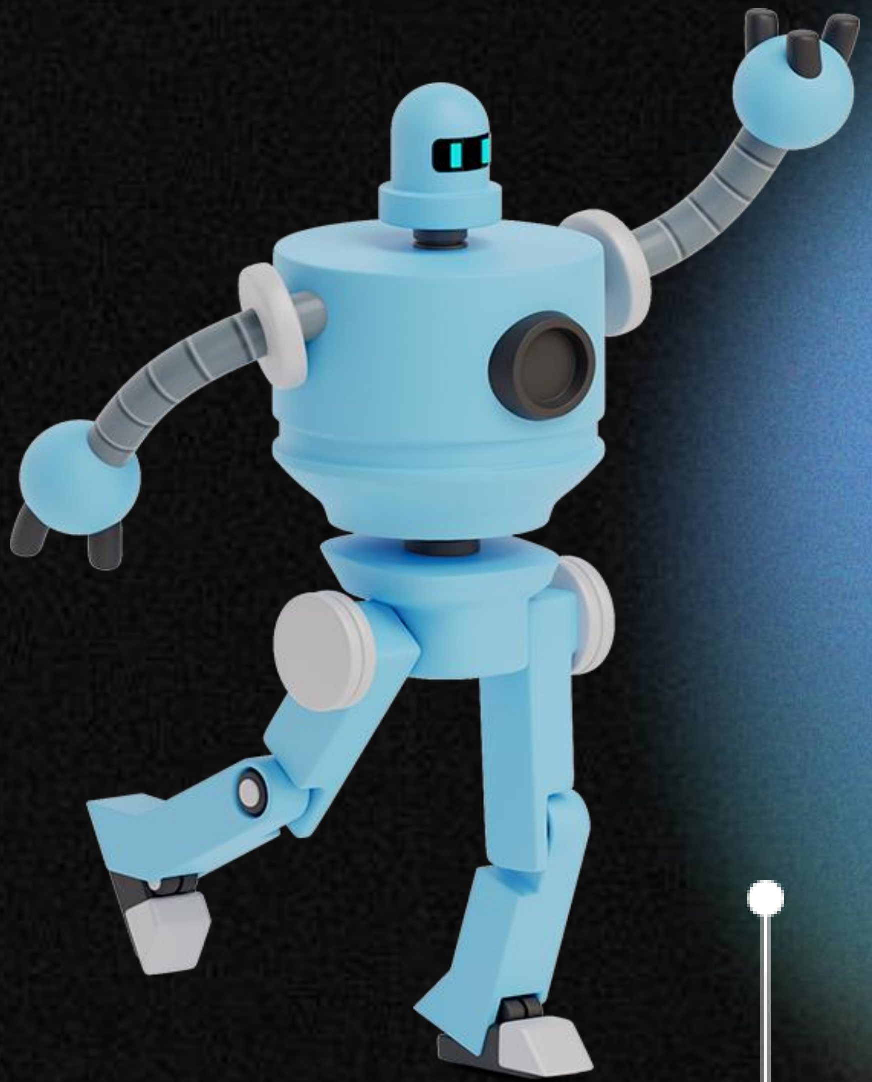
2D PRINTER



Recruitment project K23
Group 5

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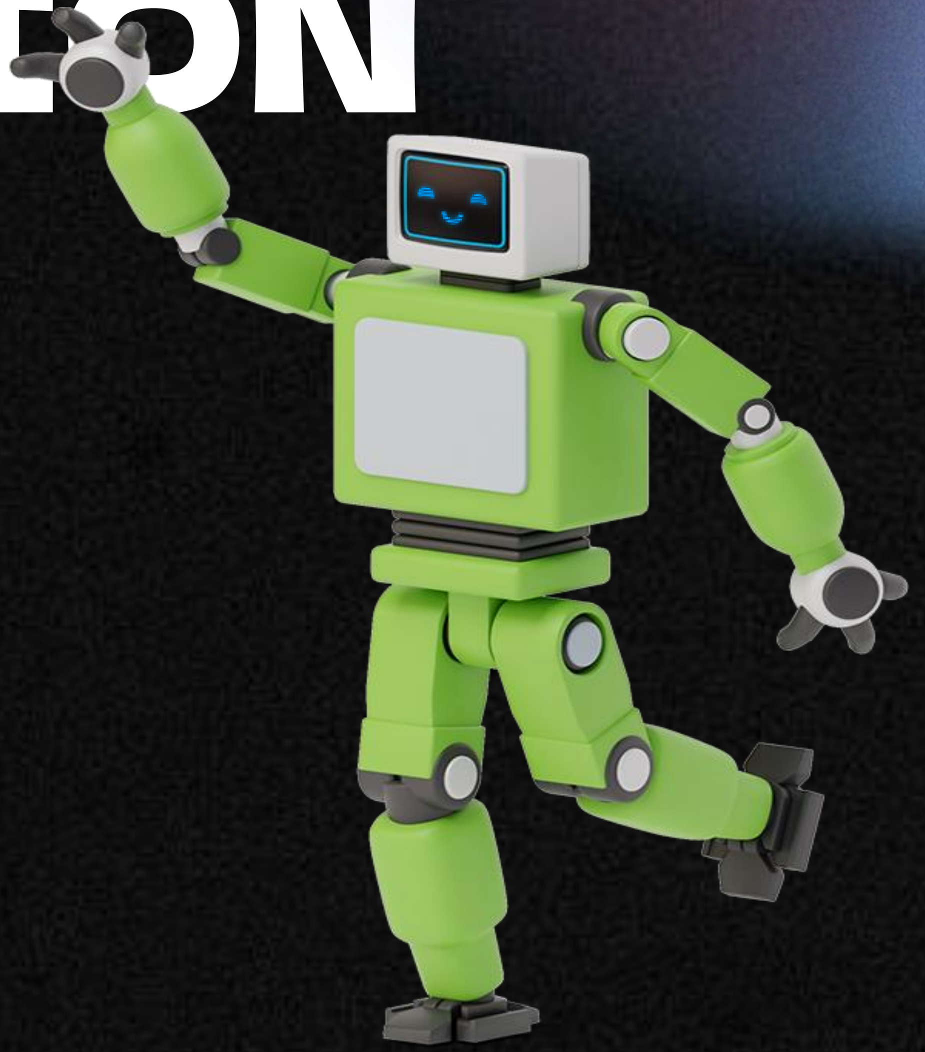
2D PRINTER

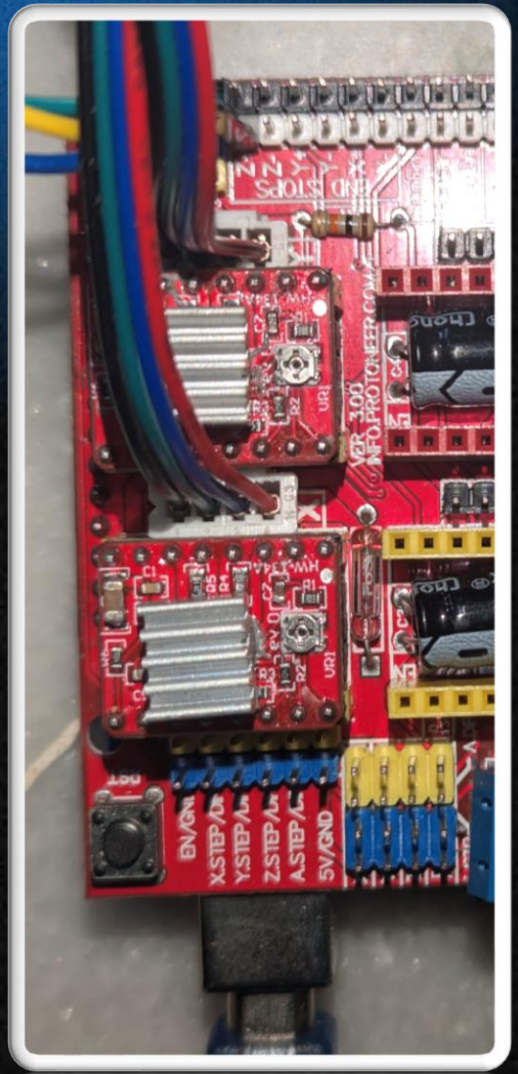
INTRODUCTION

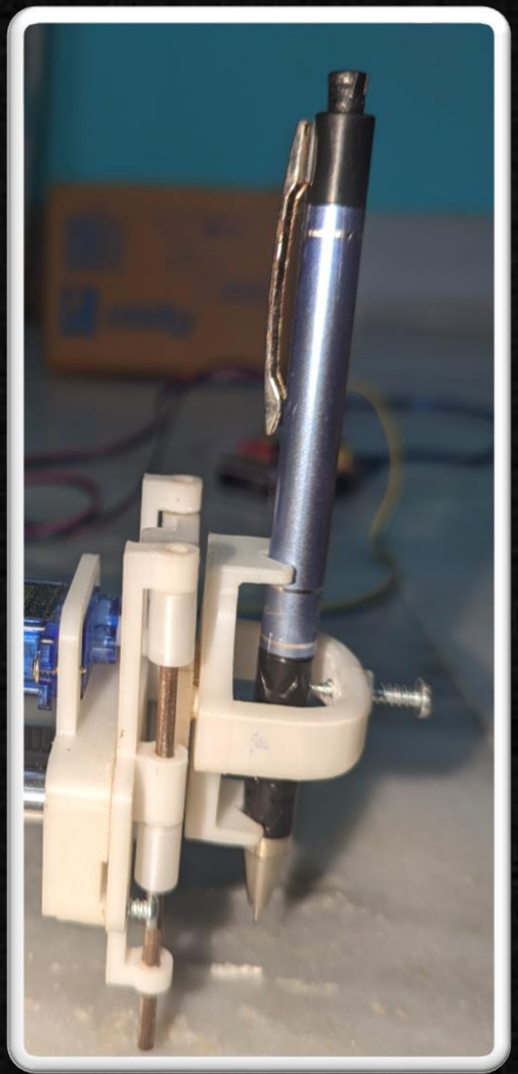
01.

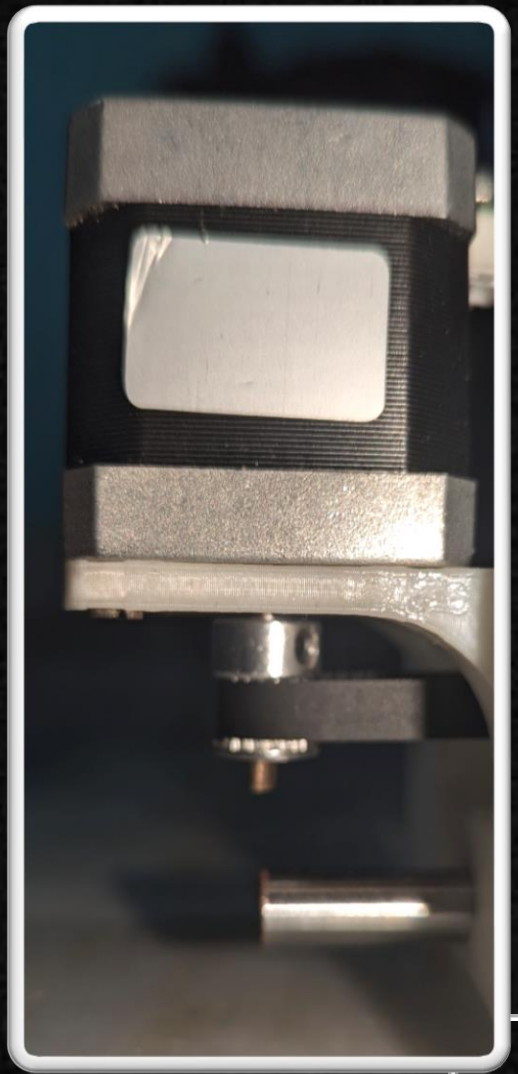


2D PRINTER :

Also known as 2D plotter, it prints text, graphics or diagrams onto a 2D surface with a plotting tool like pen.







OUR MODEL

Single belt design

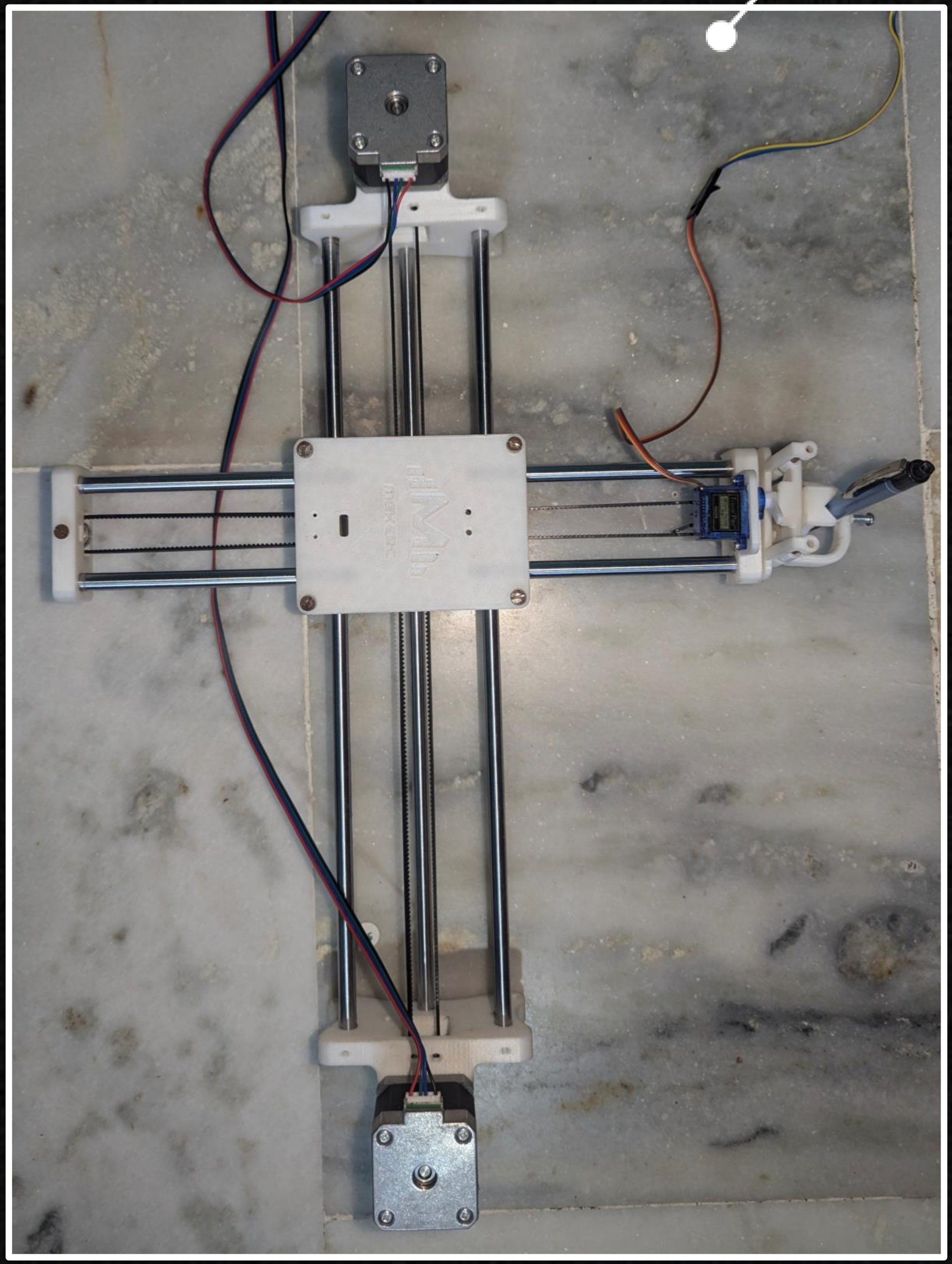
Rather than using 2 belts for 2 axes, we've used a single belt which is fixed at an end.

Position of motors

2 NEMA motors are fixed at 2 opposite ends and one servo motor is used for pen lift mechanism.

Sturdy

3D printed parts have been used for better fitting of all the components. Rods provide a sturdy frame.



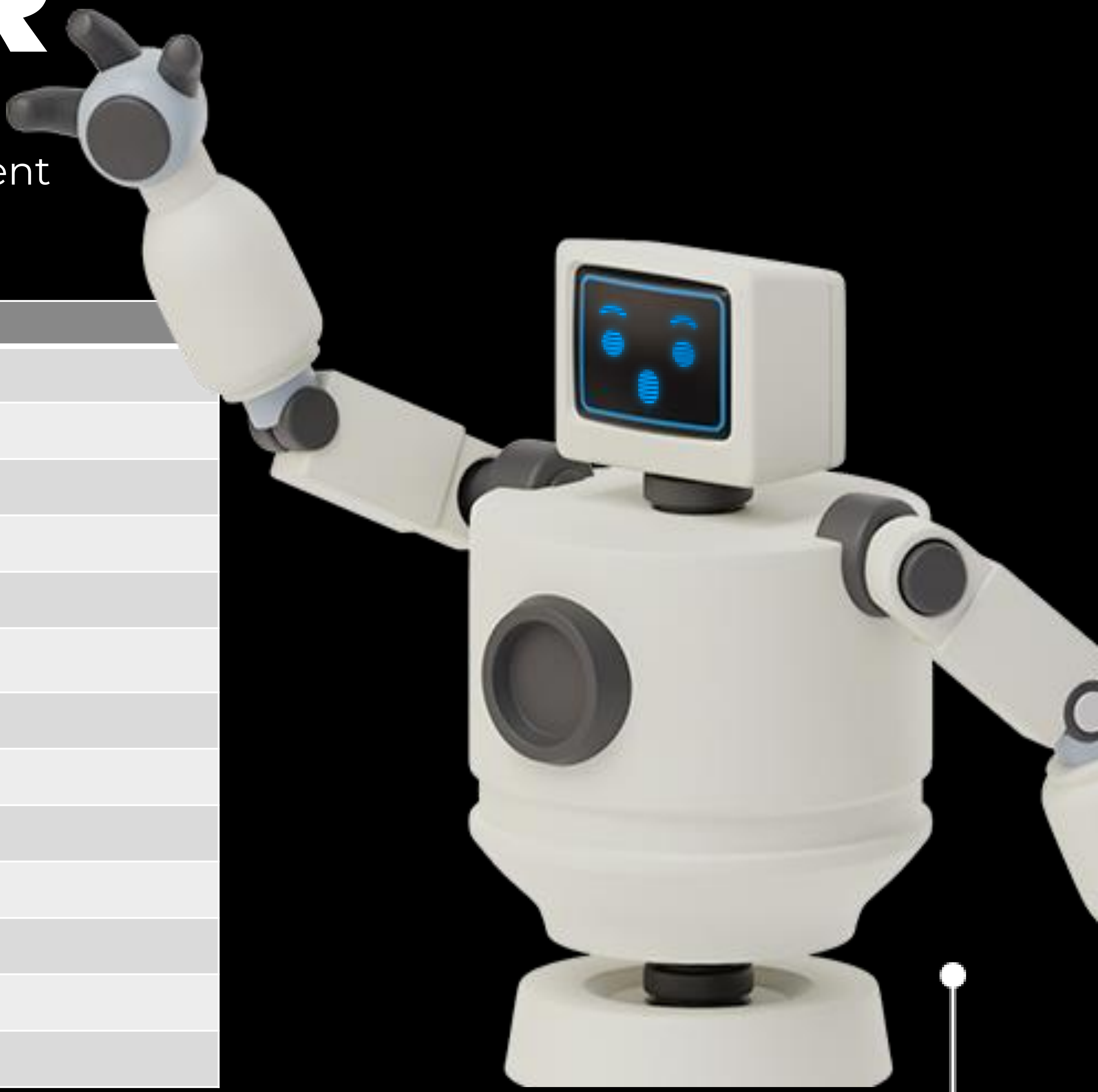
02.

BUDGET

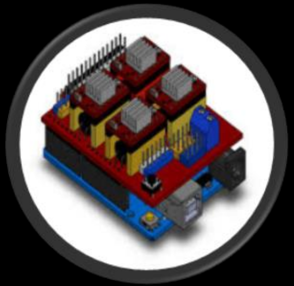
6,200 INR

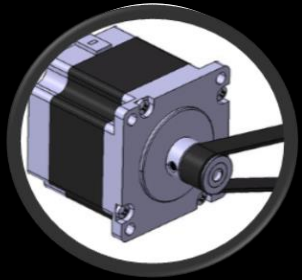
This is the amount of money we have spent on this project, following are the details :

PARTS	PRICE
SERVO MOTOR	150
GT2 BELT + PULLEY X 2	600
LM8UU LINEAR BEARING X 8	312
ARDUINO UNO	400
CNC SHIELD	105
RADIAL BALL BEARING 624ZZ X 8	112
A4988 MOTOR DRIVER X 2	100
NEMA 17 STEPPER MOTORS X 2	800
JUMPER WIRES AND MINI JUMPER CAPS 40 EACH	200
12V 2A DC POWER SUPPLY	200
3D PRINTED PARTS	2000
RODS	650
MISCELLANEOUS	500



03. WORKING PRINCIPLE





Arduino receives codes from the laptop.

CNC commands motors to rotate.

Motors move the belt. Servo rotates whenever pen lift and drop is required.

The belt applies tension at the fixed end and makes the machine move.

IMAGE PROCESSING

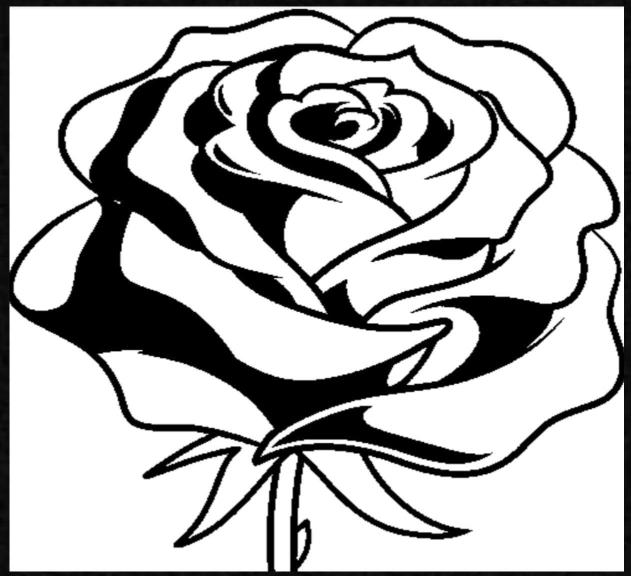
04.

Here we convert desired image into a format suitable to generate Gcode.

The processes involved are :

Imported

Greyscale



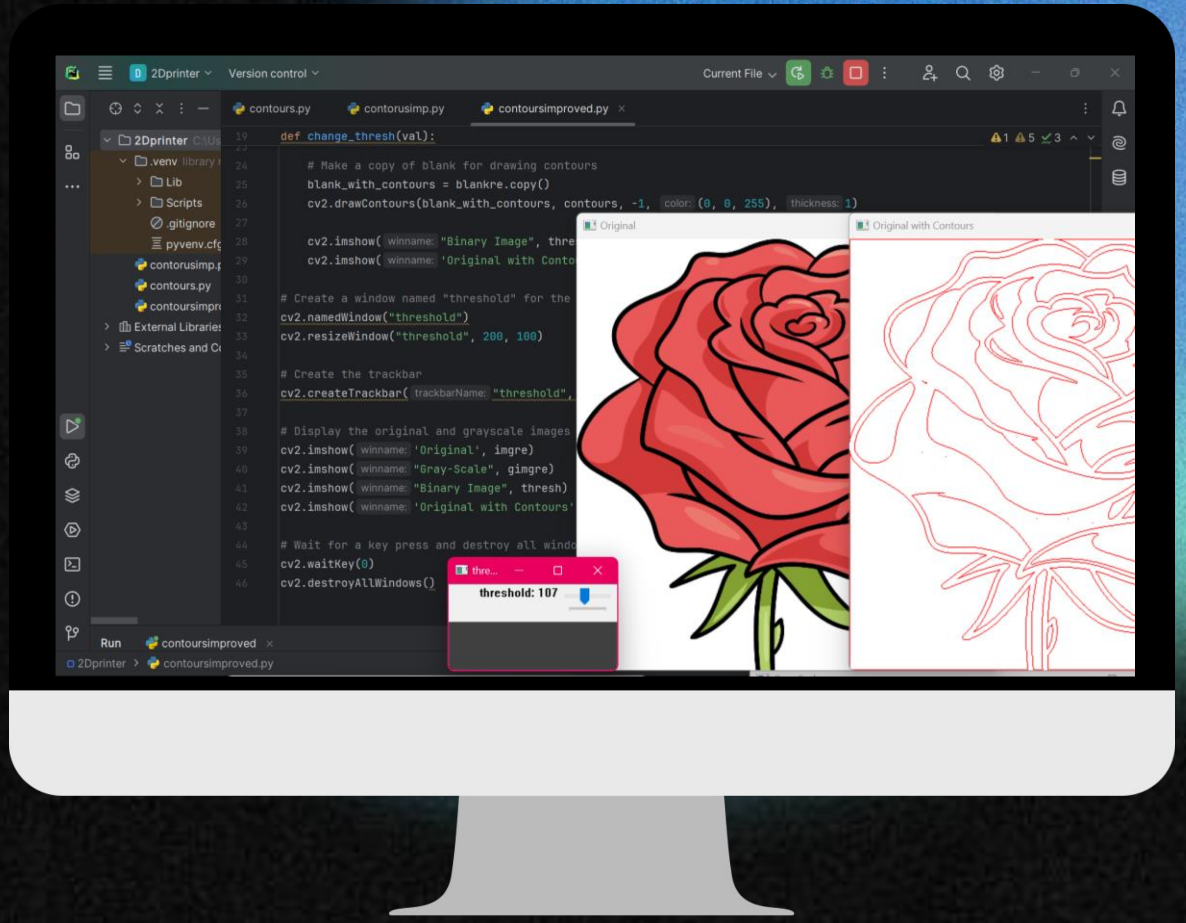
Binary

Contours

Now this image is ready to be fed to the software and generate Gcode.

Image processing, HOW?

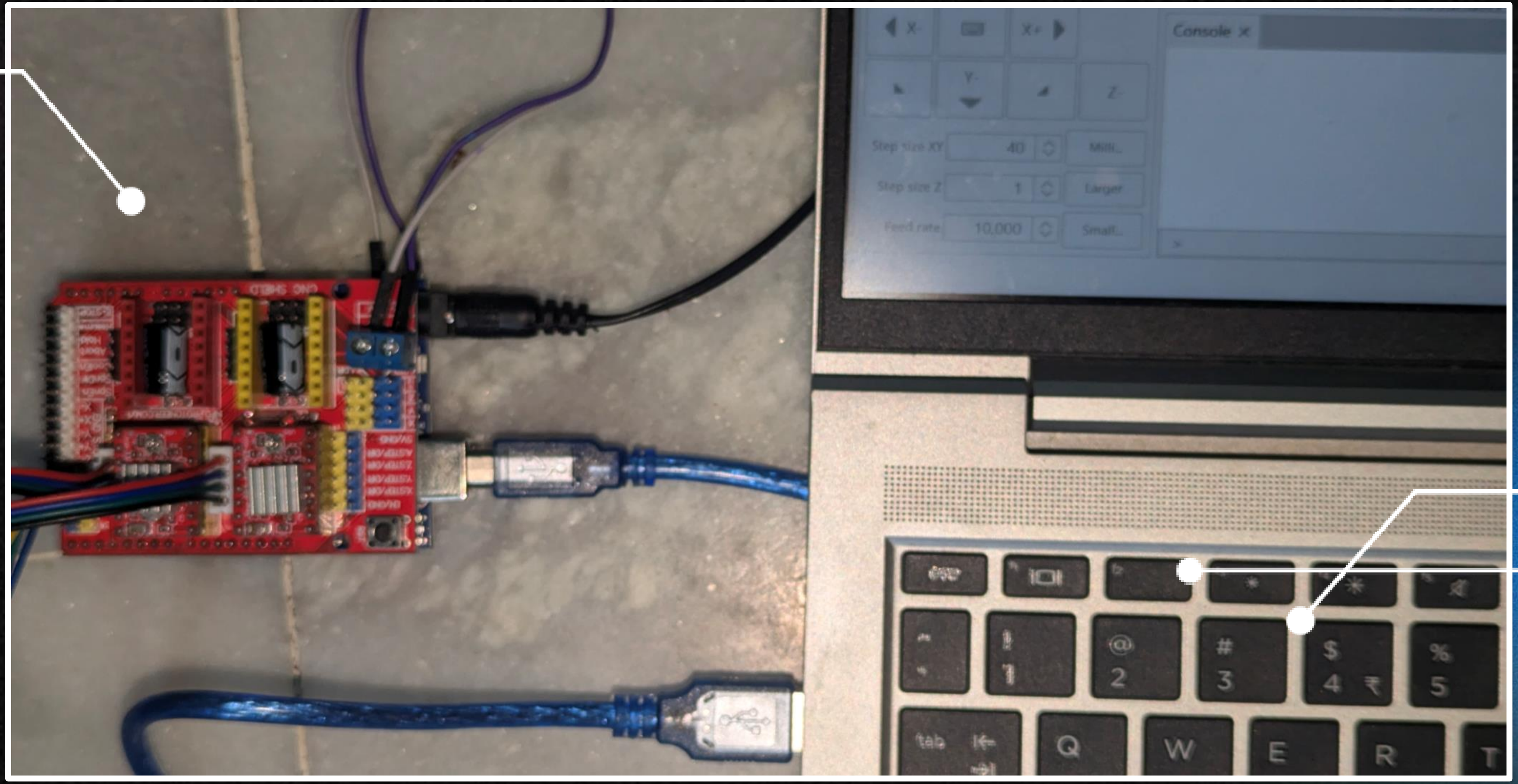
Image processing shown in the previous slide can be achieved by codes which we've written in OpenCV python. Our code identifies contours of the raw image and draws them. We have also added a trackbar which facilitates the amount of threshold we're applying.
Code can be accessed here : https://github.com/hameeddmd/2d-Printer/tree/main



Alternatively, we can use a software to do all of this and generate the required Gcode. We're using inkscape 0.48.5 and the MI-GRBL extension for this.

05.

SOFTWARE-HARDWARE INTEGRATION



HARDWARE :

Connect motor drivers and motors according to the wiring diagram.

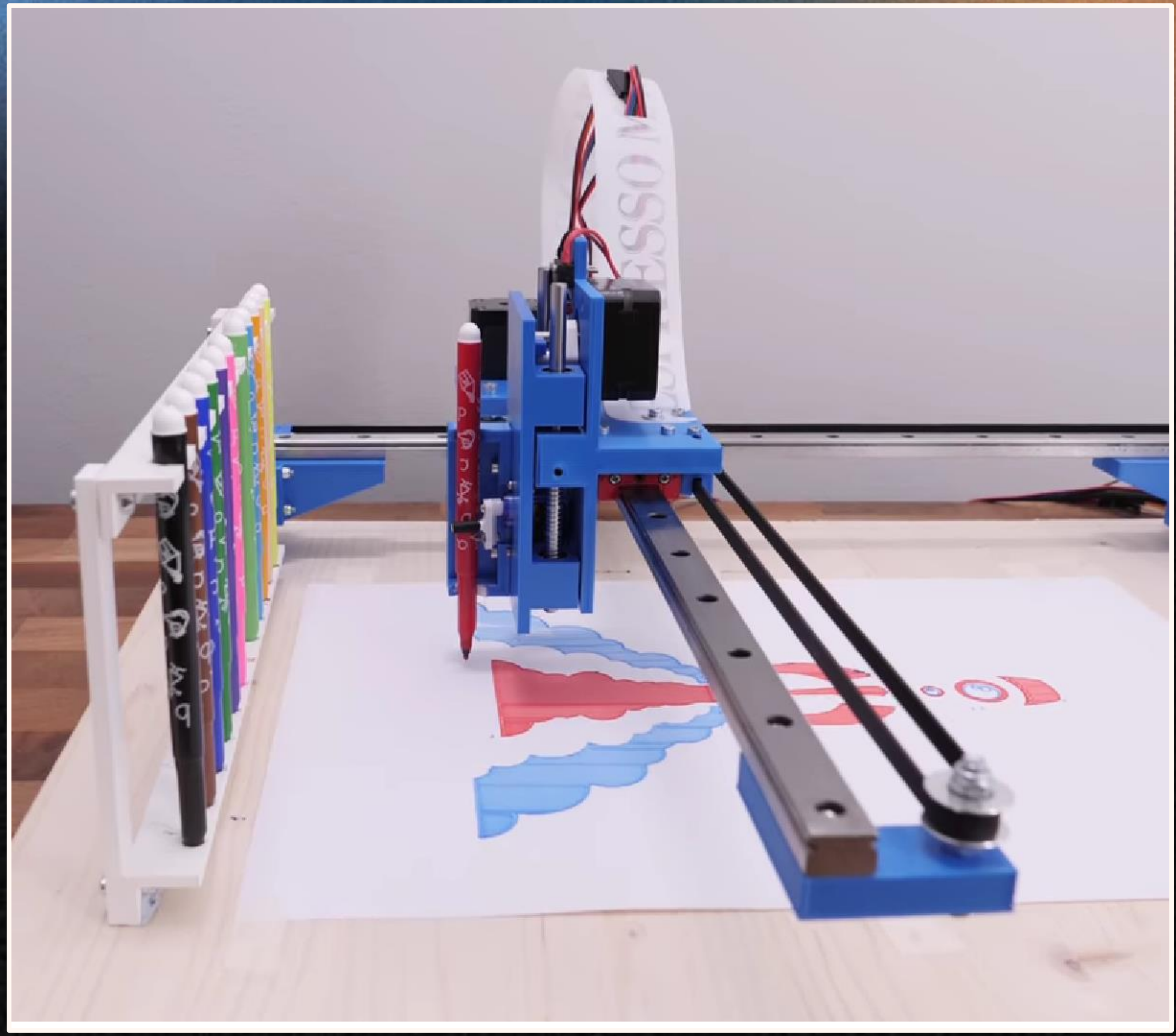
Place cnc on the Arduino board.

Connect Arduino to laptop and upload firmware.

SOFTWARE :

Install GRBL firmware on Arduino, now Arduino is ready to accept Gcode commands. Generate Gcode using inkscape and MI GRBL extension.

Next we send Gcode using Universal Gcode sender.



FUTURE ASPECTS

A pen changing tool can be implemented to have different colors and textures in the drawing.

Pressure and angle can be varied for a more accurate drawing.

THANK YOU !

OUR TEAM :

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Group Mentor (K22)
Mechanical Head at
Robolution

Shantanu

Group Member
Mechanical Subteam
Intern

DMD Hameed

Group Member
Mechanical Subteam
Intern

Jayanth Shetty

Group Member
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Intern

Visit our project at github :
https://github.com/hameeddmd/2d-Printer/tree/main